SAISREE KRISHNARAJENDRAN

Senior Software Engineer | Lead Full-Stack Developer (Angular & .NET Core)
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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 9 years designing, architecting, and delivering enterprise-scale web applications across construction technology and healthcare. Owns end-to-end architecture for 20+ modules spanning C# / .NET Core Web API services and Angular (v5-20) clients inside a mono-repo platform serving 10,000+ users and processing take-off datasets of hundreds of thousands of records. Architected distributed transaction workflows using MassTransit Saga Orchestration and State Machines over Azure Service Bus, and delivered repeated performance engineering wins including a 94% grid reload reduction and a 12-minute to sub-1-minute load-time reduction. Leads proof-of-concept and spike research that sets platform-wide technology direction, reviews peer code, and mentors junior engineers within Agile/Scrum teams shipping quarterly production releases to global onshore stakeholders. Microsoft Certified in Copilot for Microsoft 365, applying AI-assisted development to day-to-day engineering work.

CORE COMPETENCIES

Backend Engineering: C# | .NET Framework | .NET Core | .NET 8 | ASP.NET Core 8.0 | Web API | RESTful API Design | Entity Framework | LINQ | Microservices | SignalR | MassTransit

Distributed Systems: Saga Orchestration | MassTransit State Machines | Routing Slips | Distributed Transactions | Compensating Transactions & Rollback | Event-Driven Architecture | Real-Time Messaging | Asynchronous Processing | Message-Driven Workflows

Databases & Cloud: MongoDB | MongoDB Aggregation Pipelines | MS SQL Server | Microsoft Azure | Azure Service Bus | Data Modeling | Query Optimization | MongoDB Compass

Frontend Engineering: Angular 5-20 (incl. Angular 18, Angular 20) | TypeScript | NgRx | Redux | RxJS | Web Workers | Angular Material | HTML5 | CSS3 | Dynamic Forms | OPFS | PouchDB | IndexedDB | Grid Performance Optimization

Architecture & Design: Mono-repo Architecture | System Design | HLD & LLD | Data Structures & Algorithms | Performance Optimization | Multi-Threaded Background Processing | Client-Side Storage Architecture | Offline-First Design | Technology Evaluation & POC Leadership

Quality & Delivery: Code Review | Unit & Integration Testing (95%+ coverage) | Defect Triage & Hotfix Management | Deployment Management | Agile/Scrum | Sprint Planning | Estimation | Backlog Grooming | Requirement Gathering

Leadership: Technical Mentoring | Scrum Master | Stakeholder & Architecture Reviews | Cross-Team Collaboration | Product Demos | Onshore/Offshore Coordination

Tooling: Git | Visual Studio | VS Code | Postman | GitHub Copilot | Jira | Miro | Git Kraken

PROFESSIONAL EXPERIENCE

TRIMBLE INFORMATION TECHNOLOGIES

Senior Software Engineer (Software Engineer II) | Chennai, India | Feb 2020 - Present
Joined as Software Engineer I; promoted to Software Engineer II.

- Own end-to-end design, architecture, and delivery of 20+ full-stack modules spanning C# / .NET Core Web API services and Angular clients within a mono-repo construction-estimation platform serving 10,000+ users and processing take-off datasets of hundreds of thousands of records.
- Architected a distributed transaction system using MassTransit State Machines and Saga Orchestration within a microservices architecture over Azure Service Bus, streamlining complex multi-service workflows and synchronizing real-time state updates to clients via SignalR.
- Integrated MassTransit Routing Slips to automate compensation logic and rollbacks across multi-stage processes, ensuring 100% system recoverability from partial failures.
- Cut grid reload time by 94%, from 150 seconds to 9 seconds, by engineering a custom SHA-256 checksumming strategy for change detection.
- Designed and optimized MongoDB aggregation pipelines to transform, group, and summarize large take-off datasets powering estimation and reporting workflows.
- Reduced a critical load-time bottleneck from 12 minutes to under 1 minute by architecting an OPFS-based client-side data layer with Web Workers, offloading real-time calculation across hundreds of thousands of records to background threads and keeping the UI responsive.

- Led proof-of-concept and spike research that selected the platform's core technology stack - ASP.NET 8.0, MongoDB, MassTransit with Azure Service Bus, SignalR, Angular 20, NgRx, OPFS, and PouchDB/IndexedDB - with each decision adopted platform-wide; delivered Angular dynamic forms and cross-instance browser messaging into production, and prototyped AI agent-based automated resolution of pull-request review comments.
- Review and approve 4-5 peer pull requests per sprint and manage deployments for a team of 8-10 engineers on a biweekly sprint cadence and quarterly production release schedule, sustaining 90% on-time delivery and 95%+ unit test coverage across owned features.
- Mentor 4-5 junior developers and serve as primary technical liaison in HLD/LLD, architecture, and stakeholder reviews with Product Owners and Architects; contributed to a cross-team Advisory Group resolving cross-cutting technical concerns, and stepped into Scrum Master duties for sprint delivery alongside full development responsibilities.

ATOS | SYNTEL PRIVATE LIMITED

Associate Consultant | Pune, India | Aug 2017 - Feb 2020

- Redefined the application architecture of a physician performance-evaluation platform serving 10 healthcare facilities and approximately 1,000 physicians, improving scalability and maintainability across the C# / .NET, Web API, Entity Framework, and SQL Server stack.
- Cut page load times from 5-20 minutes to under 1 minute through architectural redesign and lazy-loading implementation, sharply reducing resource consumption.
- Built and deployed full-stack features using C#, .NET Framework, Entity Framework, LINQ, Web API, and Angular 5/6 with Angular Material.

KEY PROJECTS

Live Count - 3D Quantity Take-off | Trimble | Feb 2023 - Present
Next-generation 2D/3D building construction estimating platform; 3D estimation capability preparing for launch.
Stack: C# .NET, ASP.NET 8.0, Web API, MongoDB, MassTransit, Azure Service Bus, SignalR, Angular 20, TypeScript, NgRx, RxJS, Web Workers, OPFS

Estimation Construct | Trimble | Feb 2020 - Jan 2023
Web-based construction estimating product, including MEP estimation.
Stack: C# .NET, ASP.NET, Web API, SQL Server, Angular 12, TypeScript, NgRx, RxJS

Professional Practice Evaluation (PPE) | Atos | Syntel | Jul 2018 - Jan 2020
Healthcare platform evaluating and improving physician performance across 10 facilities and ~1,000 physicians.
Stack: C#, SQL Server, Web API, Entity Framework, LINQ, Angular 6, Angular Material

Physician Link | Atos | Syntel | Oct 2017 - Jun 2018
Web application enabling physicians to manage professional and practice details.
Stack: C#, SQL Server, Web API, Entity Framework, LINQ, Angular 5, Angular Material

LEADERSHIP AND ADDITIONAL CONTRIBUTIONS

- Member, Glint Survey Advisory Board - internal committee shaping employee engagement action planning.
- Represented the engineering team in cross-team technical forums and delivered multiple product demos and Quarterly Engineering Review (QER) presentations across divisions.
- Participated in multiple internal hackathons and carried concepts into day-to-day production engineering practice.
- Partnered with teams across global regions, serving as an embedded temporary developer to unblock complex technical solutions.
- Contributed to a new customer win for the estimation software through technical demonstration and solution support.

CERTIFICATIONS

- Microsoft Certified: Copilot for Microsoft 365 Applied Skills

EDUCATION

B.E., Computer Science Engineering
Jansons Institute of Technology, Coimbatore | May 2017